Can Pretext-Based Self-Supervised Learning Be Boosted by Downstream Data? A Theoretical Analysis

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Abstract

Pretext-based self-supervised learning aims to learn the semantic representation via a handcrafted pretext task over unlabeled data and then use the learned representation for downstream prediction tasks. It is proved that pretext-based self-supervised learning can effectively reduce the sample complexity of downstream tasks under Conditional Independence (CI) between the components of the pretext task conditional on the downstream label [15]. However, the downstream sample complexity will get much worse if the CI condition does not hold. One interesting question is whether we can make the CI condition hold by using downstream data to refine the unlabeled data to boost self-supervised learning. At first glance, one might think that seeing downstream data in advance would always boost the downstream performance. However, we show that it is not intuitively true and point out that in some cases, it will hurt the final performance instead. In particular, we prove both model-free and model-dependent lower bounds of the number of downstream samples used for data refinement. Moreover, we conduct several experiments on both synthetic and real-world datasets to verify our theoretical results.

1 Introduction

Data representations used to be learned in a supervised or semi-supervised learning way (e.g., [10, 14]). Recently, self-supervised learning has drawn massive attention for its fantastic data efficiency and generalization ability, with many state-of-the-art models following this paradigm in computer vision [3, 11], language modeling [4, 22], graph learning [21], etc. It learns data representations through self-supervised tasks, and then uses the learned representations for downstream prediction tasks. Such paradigm involves a large number of unlabeled data that are easier to access, to reduce the sample complexity of labeled data.

The renaissance of self-supervised learning began with artificially designed *pretext tasks* as self-supervised tasks, such as image colorization [28], inpainting [20], solving jigsaw puzzles [18], predicting image rotations [8], etc. Meanwhile, some experiments reveal that the representations learned from inappropriate pretext tasks may fail to transfer to downstream tasks [27, 29]. Thus, how pretext tasks will affect the performance of downstream tasks is crucial to self-supervised learning.

There are some empirical studies of the above question (e.g., [17]), however, the theoretical study is still at an early stage. A recent work [15] proves that, under conditional independence between the components of the pretext task conditional on the downstream label, the downstream task achieves minimal sample complexity. In particular, it shows that for Gaussian variables, the sample complexity of downstream tasks can be reduced to $\tilde{\mathcal{O}}(\dim(\boldsymbol{y}))$[1] when Conditional Independence (CI) $\boldsymbol{x} \perp \boldsymbol{z} \mid \boldsymbol{y}$ holds, where $\boldsymbol{x}$, $\boldsymbol{z}$, $\boldsymbol{y}$ indicates the input variable, pretext label, and downstream label, respectively. As a comparison, the sample complexity of directly using $\boldsymbol{x}$ to predict $\boldsymbol{y}$ is $\tilde{\mathcal{O}}(\dim(\boldsymbol{x}))$, where the dimension of $\boldsymbol{x}$ is supposed to be much larger than the dimension of $\boldsymbol{y}$. However, if the CI condition

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[1]We use notation $\tilde{\mathcal{O}}$ to hide log factors in this paper, and abbreviation "dim" stands for dimension.

$\boldsymbol{x} \perp \boldsymbol{z} \mid \boldsymbol{y}$ does not hold, the downstream sample complexity will increase to $\tilde{\mathcal{O}}(\dim(\boldsymbol{z}))$. This is because there is redundant mutual information between $\boldsymbol{x}$ and $\boldsymbol{z}$ which is unrelated to $\boldsymbol{y}$.

In practice, the CI condition rarely holds, and thus self-supervised learning cannot realize its full potential. An interesting question raises:

Can we make the CI condition $\boldsymbol{x} \perp \boldsymbol{z} \mid \boldsymbol{y}$ hold with the help of downstream data to boost self-supervised learning?

To formulate the above question, we introduce a learnable function f (called data processor) to refine the pretext data such that $f(\boldsymbol{x}) \perp \boldsymbol{z} \mid \boldsymbol{y}$ holds, and the downstream data (or extra data from downstream data distribution) are allowed to access to learn the processor. If such processor f exists, according to the result of [15], the downstream sample complexity can be reduced by replacing all the unlabeled data $\boldsymbol{x}$ with $f(\boldsymbol{x})$. Otherwise, self-supervised learning cannot be boosted by the downstream data (w.r.t. the order of sample complexity).

At first glance, one might think that seeing downstream data in advance would always boost downstream task performance. Furthermore, if the accessible downstream data are the extra data from downstream data distribution, then it is more likely to boost downstream task performance since there are additional data involved. However, we show that the above intuition is not always true and point out that in some cases, the downstream performance will be hurt instead. Therefore, our results validate self-supervised learning in some sense, since we prove that it is better not to use downstream data in such cases, as the standard self-supervised learning does.

Specifically, we first rigorously formulate the necessary conditions that data processor f needs to satisfy in Section 2. One is a relaxation of condition $f(\boldsymbol{x}) \perp \boldsymbol{z} \mid \boldsymbol{y}$, and the other condition ensures that $f(\boldsymbol{x})$ has enough ability to predict $\boldsymbol{y}$. We then design an ingenious loss function to learn such function f in Section 3. We demonstrate the rationality of our loss by proving that any function f minimizing the proposed loss satisfies the above criteria. Based on the loss, we theoretically study the number of downstream data required. We derive a model-free lower bound in Section 4.1 and show that involving limited downstream data will hurt downstream task performance. Furthermore, we take the model capacity of function class into account and provide a model-dependent lower bound in Section 4.3. The lower bound gets higher when the model capacity gets larger, indicating that the number of downstream samples needs to match the model capacity. To verify our theoretical results, we conduct experiments on both synthetic and real-world datasets in Section 5. We defer all the proofs in the supplementary materials.

In summary, our contributions are listed as follows:

- *Novelty*: We rigorously prove that involving the extra downstream data provably fails to boost self-supervised learning with insufficient downstream samples, which validates the standard self-supervised learning framework.
- *Technical Contributions*: We provide both model-free and model-dependent lower bounds for the sample size of extra downstream data. Moreover, we show that more complex model classes are more likely to lead the failure.
- *Experiments*: We conduct several experiments on both synthetic dataset and real-world dataset to verify our theoretical results. We observe that training with insufficient extra downstream data will significantly decrease downstream task performance.

2 Notations and Problem Formulation

Let $\boldsymbol{x} \in \mathcal{X} \subset \mathbb{R}^{d_x}$, $\boldsymbol{z} \in \mathcal{Z} \subset \mathbb{R}^{d_z}$, $\boldsymbol{y} \in \mathcal{Y} \subset \mathbb{R}^{d_y}$ denote the input variable, pretext label, and downstream label, respectively, where the data are sampled from a joint distribution $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) \sim \mathcal{P}$ and $d_x \geq d_z \geq d_y$. We use bold fonts to denote random variables in this paper. We assume $\mathbb{E}[\boldsymbol{x}] = \mathbb{E}[\boldsymbol{z}] = 0$ without loss of generality and denote $\mathrm{Cov}[\boldsymbol{x}, \boldsymbol{z}|\boldsymbol{y}] \triangleq \mathbb{E}[\boldsymbol{x}\boldsymbol{z}^\top] - \mathbb{E}[\boldsymbol{x}\boldsymbol{y}^\top](\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top])^{-1}\mathbb{E}[\boldsymbol{y}\boldsymbol{z}^\top]$ the partial covariance matrix. Unless stated otherwise, we consider the L_2-loss function in the following content. When subscript is omitted, notation $\|\cdot\|$ stands for L_2-norm or Frobenius norm for vectors and matrices.

In this paper, we follow the formulation of pretext-based self-supervised learning proposed in [15]. A representation $\psi^* \colon \mathbb{R}^{d_x} \to \mathbb{R}^{d_z}$ is first learned from the pretext task data by

$$\psi^* \triangleq \arg\min_{\psi \in \mathcal{H}} \mathbb{E} \left\| \psi(\boldsymbol{x}) - \boldsymbol{z} \right\|^2,$$

where $\mathcal{H}$ is the candidate function class (e.g., the class of multi-layer neural networks). Then a linear layer $W \in \mathbb{R}^{d_y \times d_z}$ following the representation $\psi^*(\boldsymbol{x})$ is learned via the downstream task, i.e.,

$$W^* \triangleq \arg\min_{W \in \mathbb{R}^{d_y \times d_z}} \mathbb{E} \left\| W\psi^*(\boldsymbol{x}) - \boldsymbol{y} \right\|^2.$$

Finally, we use $g(\cdot) = W^*\psi^*(\cdot)$ as the predictor for the downstream task.

In practice, one can only access finite samples both for the pretext and downstream tasks. We assume that we have n_1 pretext task samples $(x^1_{pre}, z^1_{pre}), \ldots, (x^{n_1}_{pre}, z^{n_1}_{pre})$ and n_2 downstream task samples $(x^1_{down}, y^1_{down}), \ldots, (x^{n_2}_{down}, y^{n_2}_{down})$, which are i.i.d. drawn from the joint distribution of $(\boldsymbol{x}, \boldsymbol{z})$ and $(\boldsymbol{x}, \boldsymbol{y})$, respectively. For simplicity, we express the samples as matrix pairs $(X_{pre}, Z_{pre}) \in \mathbb{R}^{d_x \times n_1} \times \mathbb{R}^{d_z \times n_1}$ and $(X_{down}, Y_{down}) \in \mathbb{R}^{d_x \times n_2} \times \mathbb{R}^{d_y \times n_2}$. Then the empirical representation can be written as

$$\widehat{\psi} = \arg\min_{\psi \in \mathcal{H}} \frac{1}{n_1} \left\| \psi(X_{pre}) - Z_{pre} \right\|^2,$$

where $\psi(X_{pre})$ denotes $[\psi(x^1_{pre}), \ldots, \psi(x^{n_1}_{pre})] \in \mathbb{R}^{d_z \times n_1}$. Similarly, the empirical linear layer learned can be written as

$$\widehat{W} = \arg\min_{W \in \mathbb{R}^{d_y \times d_z}} \frac{1}{n_2} \left\| W\widehat{\psi}(X_{down}) - Y_{down} \right\|^2,$$

and the final predictor for the downstream task is $\hat{g}(\cdot) = \widehat{W}\widehat{\psi}(\cdot)$.

The above pretext-based self-supervised learning is summarized as follows:

- *Step 1 (pretext):* Learn representation $\widehat{\psi}$ from pretext task samples (X_{pre}, Z_{pre}).
- *Step 2 (downstream):* Perform linear regression of Y_{down} on $\widehat{\psi}(X_{down})$ and get $\widehat{W}$.

Lee et al. [15] point out that under the Conditional Independence (CI) condition $\boldsymbol{x} \perp \boldsymbol{z} \mid \boldsymbol{y}$, self-supervised learning can largely reduce the downstream sample complexity. However, when CI does not hold, the downstream sample complexity gets worse. This is because there is much redundant information (irrelevant to the downstream task) involved in the representation during the learning via the pretext task. Therefore, for the downstream task, the mapping from the redundant features to the downstream label also needs to be learned, which causes a larger sample complexity.

In practice, the CI condition rarely holds, and thus pretext-based self-supervised learning cannot realize its full potential. An interesting question raises: *Can we make the CI condition hold with the help of downstream data to reduce the downstream sample complexity?* To formulate the above question, we introduce a learnable data processor f to refine the unlabeled data such that CI condition $f(\boldsymbol{x}) \perp \boldsymbol{z} \mid \boldsymbol{y}$ holds. If such f exists, then one can simply replace all the unlabeled data X with $f(X)$ in self-supervised learning to reduce the downstream sample complexity. Otherwise, the downstream sample complexity cannot be further improved (w.r.t. the order of sample complexity).

Formally, the following two criteria are essential for a meaningful processor[2]:

$$\mathrm{Cov}[f(\boldsymbol{x}), \boldsymbol{z} \mid \boldsymbol{y}] = 0, \tag{C1}$$

$$\mathbb{E} \left\| \boldsymbol{y} - W^*_{\boldsymbol{y}, f(\boldsymbol{x})} f(\boldsymbol{x}) \right\|^2 = \min_{f'} \mathbb{E} \left\| \boldsymbol{y} - W^*_{\boldsymbol{y}, f'(\boldsymbol{x})} f'(\boldsymbol{x}) \right\|^2, \tag{C2}$$

where $W^*_{\boldsymbol{y}, f(\boldsymbol{x})} \triangleq \arg\min_{W \in \mathbb{R}^{d_y \times d_f}} \mathbb{E} \left\| \boldsymbol{y} - Wf(\boldsymbol{x}) \right\|^2$ is defined as the best linear predictor of $\boldsymbol{y}$ on $f(\boldsymbol{x})$. Criterion (C1) guarantees that the processor $f(\boldsymbol{x})$ removes the redundant information which is useless for predicting $\boldsymbol{y}$. In fact, Criterion (C1) is a Conditional Uncorrelatedness (CU) condition, which is a relaxation of the CI condition. We use CU instead of CI because CU is a weaker condition than CI when proving a negative result. In addition, setting $\mathrm{Cov}[f(\boldsymbol{x}), \boldsymbol{z} \mid \boldsymbol{y}] = 0$ as the only objective

[2] Assume that $\mathbb{E}[f(\boldsymbol{x})] = 0$ for simplicity.

may lead to non-informative processors. For example, if $f(\boldsymbol{x})$ is an independently random noise, criterion $\mathrm{Cov}[f(\boldsymbol{x}), \boldsymbol{z} \mid \boldsymbol{y}] = 0$ naturally holds, but such $f(\boldsymbol{x})$ does not carry any information of $\boldsymbol{y}$ and thus it cannot be used to predict $\boldsymbol{y}$. Therefore, we need Criterion (C2) to ensure that applying function f to input variable $\boldsymbol{x}$ will not lose the information for predicting $\boldsymbol{y}$.

Since Criterion (C1) is related to the downstream label $\boldsymbol{y}$, downstream data must be used to obtain the processor f. They can be part of the downstream samples or extra samples from the downstream data distribution. The latter setting is more likely to satisfy the criteria than the former one, since there are additional data involved. We are trying to get the negative result, thus we consider the latter and harder setting. Since processor f is allowed to be non-linear, we use a linear layer following the processor $f(\boldsymbol{x})$ as the representation $\widehat{\psi}(\boldsymbol{x})$ for simplicity.

We remark that if processor f fails to satisfy Criterion (C1), according to the result of [15], the downstream sample complexity cannot be improved. If processor f fails to satisfy Criterion (C2), then $f(\boldsymbol{x})$ will not contain all the information for predicting $\boldsymbol{y}$, leading to a constant generalization error. Therefore, the performance of downstream task will be even worse.

Now, the question turns to be whether using extra downstream samples can make f satisfy both Criterion (C1) and (C2). To answer it, we first propose a provably rational loss for learning f in Section 3. Then we show that involving extra downstream data provably fails to get a processor satisfying the above criteria simultaneously in some cases in Section 4. Moreover, we conduct experiments and observe the existence of the above phenomenon shown in Section 5.

3 Learning Processor f

In this section, we focus on learning the processor f satisfying both Criterion (C1) and (C2). We first propose a carefully designed loss in Section 3.1, and then give a proof of its rationality in Section 3.2.

3.1 Loss Design

Recall that Criterion (C1) requires that $f(\boldsymbol{x})$ and $\boldsymbol{z}$ are uncorrelated conditional on $\boldsymbol{y}$. This indicates that $f(\boldsymbol{x})$ should have no ability to predict $\boldsymbol{z}$ under linear regimes. It can be measured by loss

$$\mathcal{L}_1 = -\mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} \left\| \boldsymbol{z} - W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x}) \right\|^2,$$

where W^* is defined as the best linear projection from the second subscript to the first subscript, i.e.,

$$W^*_{\boldsymbol{b},\boldsymbol{a}} \triangleq \arg\min_{W} \mathop{\mathbb{E}}_{\boldsymbol{b},\boldsymbol{a}} \left\| \boldsymbol{b} - W\boldsymbol{a} \right\|^2.$$

If $f(\boldsymbol{x})$ can fit $\boldsymbol{z}$ well, then $\mathcal{L}_1$ will be large. Thus, minimizing $\mathcal{L}_1$ tends to pull $\boldsymbol{z}$ and $f(\boldsymbol{x})$ apart.

For Criterion (C2), we need to guarantee that applying f to $\boldsymbol{x}$ does not lose the information for predicting $\boldsymbol{y}$, indicating that $f(\boldsymbol{x})$ can still fit $\boldsymbol{y}$ well. It can be formulated as

$$\mathcal{L}_2 = \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\| \boldsymbol{y} - W^*_{\boldsymbol{y},f(\boldsymbol{x})} f(\boldsymbol{x}) \right\|^2.$$

The smaller $\mathcal{L}_2$ indicates that $f(\boldsymbol{x})$ contains more information related to $\boldsymbol{y}$.

To minimize $\mathcal{L}_1$ and $\mathcal{L}_2$ simultaneously, we define the total population loss as $\lambda\mathcal{L}_1 + \mathcal{L}_2$, where $\lambda > 0$ is the penalty coefficient, i.e., for any data distribution $\mathcal{P}$,

$$\mathcal{L}(f;\mathcal{P}) \triangleq \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{z},\boldsymbol{y})\sim\mathcal{P}} \left[\left\| \boldsymbol{y} - W^*_{\boldsymbol{y},f(\boldsymbol{x})} f(\boldsymbol{x}) \right\|^2 - \lambda \left\| \boldsymbol{z} - W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x}) \right\|^2 \right]. \tag{1}$$

One might think that forcing $f(\boldsymbol{x})$ to fit $\boldsymbol{y}$ but not to fit $\boldsymbol{z}$ at the same time seems to be in conflict with each other, since $\boldsymbol{z}$ contains all the information of $\boldsymbol{y}$. In the following Section 3.2, we will give a strict proof of its rationality. The intuition of such loss design is to keep the useful information for predicting $\boldsymbol{y}$, as well as, remove the redundancy information unrelated to $\boldsymbol{y}$.

The corresponding training loss of $\mathcal{L}(f;\mathcal{P})$ can be defined as

$$\mathcal{L}_{n_1,n_0}(f;\mathcal{P}) \triangleq \frac{1}{n_0} \left\| Y_{extra} - \widetilde{W}_2 f(X_{extra}) \right\|^2 - \frac{\lambda}{n_1} \left\| Z_{pre} - \widetilde{W}_1 f(X_{pre}) \right\|^2,$$

where $\widetilde{W}_1$, $\widetilde{W}_2$ represent respectively the best empirical linear predictors for the pretext data and the downstream data, and n_1, n_0 are respectively the sample sizes of pretext data (X_{pre}, Z_{pre}) and extra downstream data (X_{extra}, Y_{extra}).

3.2 Rationality of Loss

We provide the following theorem to show the rationality of loss $\mathcal{L}(f;\mathcal{P})$, which states that by choosing a proper penalty coefficient λ, there exist a number of data distributions such that the processor f which minimizes the population loss $\mathcal{L}(f;\mathcal{P})$ satisfies Criterion (C1) and (C2).

Theorem 1 (Rationality of Loss)**.** *Assume that there exists a ground truth $f^* \in \mathcal{F}$ satisfying both Criterion (C1) and (C2), where $\mathcal{F}$ is the candidate function class. Assume that for each $f \in \mathcal{F}$, the matrix $\mathbb{E}[f(\boldsymbol{x})f^\top(\boldsymbol{x})]$ is nonsingular. Let $\mathcal{A}_\mathcal{P}$ be the set of processors that minimize the population loss under data distribution $\mathcal{P}$:*

$$\mathcal{A}_\mathcal{P} = \left\{ f : f \in \arg\min_f \mathcal{L}(f;\mathcal{P}) \right\}.$$

Let $\mathcal{B}_\mathcal{P}$ be the set of processor that satisfy the criteria, namely:

$$\mathcal{B}_\mathcal{P} = \{ f : f \text{ satisfies Criterion (C1) and Criterion (C2)} \}.$$

Then by choosing a proper parameter λ, there exist a number of population distributions $\{\mathcal{P}\}$'s such that every function in $\mathcal{A}_\mathcal{P}$ satisfies Criterion (C1) and Criterion (C2), namely, the following defined set $\mathbb{S}$ is not empty:

$$\mathbb{S} \triangleq \{\mathcal{P} : \mathcal{A}_\mathcal{P} \subset \mathcal{B}_\mathcal{P}\} \neq \phi.$$

In Theorem 1, we show that there *exist* a number of data distributions such that under those distributions, Criterion (C1) and (C2) hold when the population loss is minimized. We will show in the next section that for those data distributions, minimizing the corresponding empirical loss will fail to satisfy the criteria when the extra downstream sample is insufficient. Therefore, when we do not have much knowledge about the data distribution, it is better not to use downstream data, as the standard self-supervised learning does. Otherwise, we may get the worse downstream task performance.

4 Insufficient Downstream Samples Provably Fails

Theorem 1 has shown that we can learn the processor f by minimizing the population loss, such that f satisfies both Criterion (C1) and (C2). However, we can only access a finite number of downstream samples in practice. In this section, we will prove that even under the rational loss, one fails to learn a processor that satisfies both Criterion (C1) and (C2) with limited extra downstream data. Specifically, we first provide a model-free lower bound in Section 4.1 as a warm-up and extend the loss to a general loss in Section 4.2. We further take the structure of function class into account and give a model-dependent lower bound in Section 4.3.

4.1 Warm-Up: Model-Free Results

To better understand the role of the extra downstream samples, we consider the following loss (See Eq. (2)) which differs from the population loss only in the downstream part (i.e., infinite pretext samples). Concretely, we replace the population loss of downstream part with its empirical version.

$$\mathcal{L}_{\infty,n_0}(f,\mathcal{P}) \triangleq \frac{1}{n_0}\left\|Y_{extra} - \widetilde{W}_2 f(X_{extra})\right\|^2 - \lambda \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}}\left\|\boldsymbol{z} - W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x})\right\|^2. \tag{2}$$

We show that minimizing the above empirical loss fails to satisfy Criterion (C1) and (C2) simultaneously when $n_0 = o(d_f)$, where d_f is the dimension of $f(\boldsymbol{x})$.

Theorem 2 (Model-Free Lower Bound)**.** *Assume that there exists a ground truth $f^* \in \mathcal{F}$ satisfying both Criterion (C1) and (C2), where $\mathcal{F}$ is the candidate function class. And assume that there exist function $f_1 \in \mathcal{F}$ and $f_2 \in \mathcal{F}$ such that covariance $\mathrm{Cov}[f_1(\boldsymbol{x}), \boldsymbol{y}] \neq 0$ and $\mathrm{Cov}[f_2(\boldsymbol{x}), \boldsymbol{z}] = 0$. Assume that for any $f_2 \in \mathcal{F}$, the matrix $\mathbb{E}[f(\boldsymbol{x})f^\top(\boldsymbol{x})]$ is nonsingular. Let $\mathcal{A}'_\mathcal{P}$ be the set of processor*

that minimizes the training loss with $n_0 = o(d_f)$ downstream samples and infinite pretext samples, where d_f is the dimension of $f(\boldsymbol{x})$,

$$\mathcal{A}'_{\mathcal{P}} = \left\{ f : f \in \arg\min_f \mathcal{L}_{\infty,n_0}(f;\mathcal{P}) \right\}.$$

And let $\mathcal{B}_{\mathcal{P}}, \mathbb{S}$ be defined as in Theorem 1. Then there exists *a distribution $\mathcal{P}^0 \in \mathbb{S}$ (which means $\mathcal{A}_{\mathcal{P}^0} \subset \mathcal{B}_{\mathcal{P}^0}$), such that all the function that minimizes the loss $\mathcal{L}_{\infty,n_0}$ cannot meet Criterion* (C1) *and* (C2) *simultaneously, namely:*

$$\mathcal{A}'_{\mathcal{P}^0} \cap \mathcal{B}_{\mathcal{P}^0} = \phi.$$

Here $\mathrm{Cov}[f_1(\boldsymbol{x}), \boldsymbol{y}] \neq 0$ assumes that the function class of f is meaningful such that f has ability to predict y, while $\mathrm{Cov}[f_2(\boldsymbol{x}), \boldsymbol{z}] = 0$ could be easily satisfied when the function class is chosen independently with the pretext task. Theorem 2 indicates that under mild assumptions, to train a function f satisfying both Criterion (C1) and (C2), we need at least $\Omega(d_f)$ labeled downstream samples, although unlimited unlabeled data can be accessed. One can expect the larger downstream sample size requirement with finite unlabeled data.

Remark 1. When $n_0 = o(d_f)$, even if we have infinite pretext data (X_{pre}, Z_{pre}), the criteria cannot be satisfied, leading to a constant generalization error. This means that the downstream performance will get worse, even if we use infinite data (X_{down}, Y_{down}) for downstream training. Therefore, one can conclude that the failure is due to the lack of extra downstream data (X_{extra}, Y_{extra}).

4.2 Model-Free Results for General Loss

Theorem 2 is derived based on the L_2 loss that we design in Section 3.1. One may concern the generality. In fact, we can extend the loss in Theorem 2 to a more general one to strengthen our result, and show that the failure of limited downstream samples is not caused by a specific loss. Inspired by the previously designed L_2 loss, we define the *general loss* as

$$\overline{\mathcal{L}}\left(f, \lambda;\ g_2, \rho_2, g_1, \rho_1\right) = \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} g_2(\rho_2(\boldsymbol{y}, W^*_{\boldsymbol{y},f(\boldsymbol{x})} f(\boldsymbol{x}))) - \lambda \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} g_1(\rho_1(\boldsymbol{z}, W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x}))).$$

Here g_1 and g_2 are strictly increasing functions over $[0, \infty)$, λ is a positive penalty coefficient. Mappings $\rho_1 \colon \mathbb{R}^{d_z} \times \mathbb{R}^{d_z} \to \mathbb{R}_{\geq 0}$ and $\rho_2 \colon \mathbb{R}^{d_y} \times \mathbb{R}^{d_y} \to \mathbb{R}_{\geq 0}$ are distance metrics, and ρ_1 is consistent with ρ_2 in subspace $\mathbb{R}^{d_y} \times \mathbb{R}^{d_y}$, i.e., for all $a, b \in \mathbb{R}^{d_y}$, $\rho_1((a, \vec{0}_{d_z - d_y}), (b, \vec{0}_{d_z - d_y})) = \rho_2(a, b)$ holds. We abuse $W^*_{\boldsymbol{y},f(\boldsymbol{x})}$ and $W^*_{\boldsymbol{z},f(\boldsymbol{x})}$ as the best linear predictors which minimize the corresponding loss $\mathbb{E}\, g_2(\rho_2(\boldsymbol{y}, W f(\boldsymbol{x})))$ and $\mathbb{E}\, g_1(\rho_1(\boldsymbol{z}, W f(\boldsymbol{x})))$, respectively.

It is not hard to verify that the loss $\mathcal{L}(f, \mathcal{P})$ defined in Eq. (1) is a special case of the above general loss, by setting ρ_2, ρ_1 to be Euclidean distances and g_2, g_1 the square functions. The corresponding training loss with infinite pretext data is

$$\overline{\mathcal{L}}_{\infty,n_0}(f;\mathcal{P}) = \frac{1}{n_0} \sum_{i \in [n_0]} g_2\left(\rho_2\left(y^i_{extra}, \widetilde{W}_1 f(x^i_{extra})\right)\right) - \lambda \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} g_1\left(\rho_1\left(\boldsymbol{z}, W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x})\right)\right),$$

where $\widetilde{W}_1$ is the best empirical linear predictor for the downstream data.

Theorem 3 (Lower Bound for General Loss)**.** *Assume there exists a function f such that $f(\boldsymbol{x}) \perp \boldsymbol{z}$ holds, and assume $f(\boldsymbol{x}) \perp \boldsymbol{z}$ holds if and only if $\mathbb{E}\, g_1(\rho_1(\boldsymbol{z}, W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x})))$ reaches the maximum. Assume that $g_2 {g_1}^{-1}$ is convex and non-decreasing, and assume there exists a linear transformation $V \in \mathbb{R}^{d_z \times d_z}$ such that*

$$\mathbb{E}[\rho_1(\bar{\boldsymbol{y}}, V\boldsymbol{z})] < \frac{1}{M} g_2 {g_1}^{-1} \max_f \mathbb{E}\left[g_1 \rho_1\left(W^*_{V\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x}), V\boldsymbol{z}\right)\right] - \frac{1}{M} \min_f \mathbb{E}\left[g_2 \rho_2\left(W^*_{\boldsymbol{y},f(\boldsymbol{x})} f(\boldsymbol{x}), \boldsymbol{y}\right)\right],$$

where $M \geq {g_2}'(x), \forall x \in [0, \sup_{\boldsymbol{x},\boldsymbol{y}} \rho_2(W^*_{\boldsymbol{y},\hat{f}(\boldsymbol{x})} \hat{f}(\boldsymbol{x}), \boldsymbol{y}) + \sup_{\boldsymbol{y},\boldsymbol{z}} \rho_1(\bar{\boldsymbol{y}}, V\boldsymbol{z})]$, and $\bar{\boldsymbol{y}} = (\boldsymbol{y}, \vec{0}_{d_z - d_y})$ is the augmented vector of $\boldsymbol{y}$. Let $\mathcal{A}^g_{\mathcal{P}}$ be the set of processor that minimize the general training loss with $n_0 = o(d_f)$ downstream samples and infinite pretext samples, where d_f is the dimension of $f(\boldsymbol{x})$,

$$\mathcal{A}^g_{\mathcal{P}} = \left\{ f : f \in \arg\min_f \overline{\mathcal{L}}_{\infty,n_0}(f;\mathcal{P}) \right\}.$$

Let $\mathcal{B}_{\mathcal{P}}$ be defined as in Theorem 1. Then there exists *a distribution $\mathcal{P}^0$ such that all the function that minimizes the loss $\overline{\mathcal{L}}_{\infty,n_0}(f;\mathcal{P})$ cannot meet Criterion* (C1) *and* (C2) *simultaneously, namely:*

$$\mathcal{A}^g_{\mathcal{P}^0} \cap \mathcal{B}_{\mathcal{P}^0} = \phi.$$

4.3 Model-Dependent Results

The results stated in Section 4.1 hold without any assumption of model structures (i.e., the function class of f). One may wonder whether we can take the model structures into account to get the fine-grained result. The answer is yes. Before providing the main theorem, we introduce a measure of *model capacity* at first, which is defined as follows.

Definition 1 (Model Capacity)**.** *Define the model capacity $\mathcal{M}(\mathcal{F}, \mathcal{L})$ of function class $\mathcal{F}$ with respect to loss function $\mathcal{L}$ as*

$$\mathcal{M}(\mathcal{F}, \mathcal{L}) = \sup \left\{ n : \forall \mathcal{D}, \inf_{f \in \mathcal{F}} \sup_{(X,Y) \in \mathcal{D}^n} \mathcal{L}(f(X), Y) = 0 \right\},$$

where $\mathcal{D}$ is the data distribution, and X, Y are data vectors, each of which consists of n samples.

Intuitively, model capacity measures how well the function class fits the noise under finite samples. It is similar to the VC dimension under regression settings. In the following Lemma 4.1, we provide the lower bounds of the model capacity for linear models and neural networks.

Lemma 4.1 (Linear Function's Model Capacity)**.** *Let $\mathcal{J} = \{g : g(\boldsymbol{x}) = w^\top \boldsymbol{x}, \boldsymbol{x} \in \mathbb{R}^d, w \in \mathbb{R}^{k \times d}\}$ be the class of linear functions, its model capacity satisfies*

$$\mathcal{M}(\mathcal{J}, \mathcal{L}) \geq d.$$

Let $\mathcal{K}$ be the class of two-layer neural networks with k neurons, its model capacity satisfies

$$\mathcal{M}(\mathcal{K}, \mathcal{L}) \geq k/4.$$

Let $\mathcal{K}_m$ be the class of multi-layer neural networks with L layers and k_i neurons in each layer i, its model capacity satisfies

$$\mathcal{M}(\mathcal{K}_m, \mathcal{L}) \geq \min_{i \in [L]} k_i/4.$$

We next provide the model-dependent lower bound in Theorem 4:

Theorem 4 (Model-Dependent Lower Bound)**.** *Assume that the function class of the processor can be decomposed as $\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2$, where there exists a function $f_0 \in \mathcal{F}$ such that $\mathrm{Cov}[f_0(\boldsymbol{x}), \boldsymbol{y}] \neq 0$ and a function $f_2 \in \mathcal{F}_2$ such that $f_2(\boldsymbol{x}) \perp \boldsymbol{z}$. And assume matrix $\mathbb{E}[f(\boldsymbol{x}) f^\top(\boldsymbol{x})]$ is nonsingular for all $f \in \mathcal{F}$. Let $\mathcal{A}'_{\mathcal{P}}$ be the set of processor that minimize the training loss with $n_0 = o(\mathcal{M}(\mathcal{F}_1, \mathcal{L}))$ downstream samples and infinite pretext samples, where d_f is the dimension of $f(\boldsymbol{x})$,*

$$\mathcal{A}'_{\mathcal{P}} = \left\{ f : f \in \arg\min_f \mathcal{L}_{\infty, n_0}(f; \mathcal{P}) \right\}.$$

And let $\mathcal{B}_{\mathcal{P}}, \mathbb{S}$ be defined as in Theorem 1. Then there exists *a distribution $\mathcal{P}^0 \in \mathbb{S}$ (which means $\mathcal{A}_{\mathcal{P}^0} \subset \mathcal{B}_{\mathcal{P}^0}$), such that all the function that minimizes the loss $\mathcal{L}_{\infty, n_0}$ cannot meet Criterion* (C1) *and* (C2) *simultaneously, namely:*

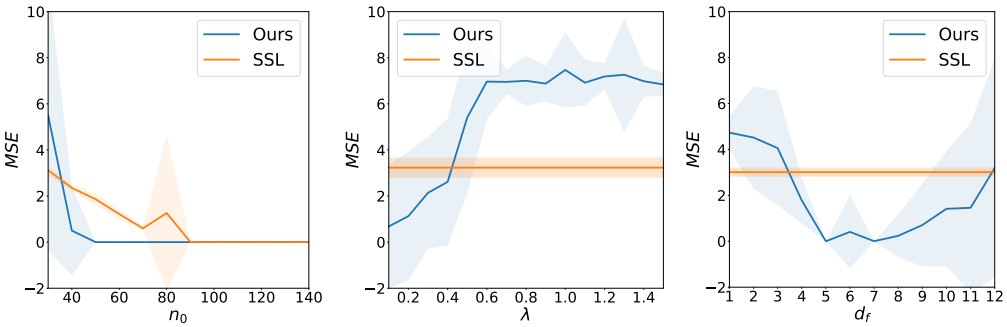
$$\mathcal{A}'_{\mathcal{P}^0} \cap \mathcal{B}_{\mathcal{P}^0} = \phi.$$

Intuitively, Theorem 4 shows that with a more complex function class to train processor, the lower bound for downstream samples is higher. Note that the model capacity is infinity for the neural network with infinite width. Therefore, the procedure provably fails due to the large model capacity.

Remark 2. The model dependent bound (Theorem 4) and the model-free bound (Theorem 2) focus on the processor training and the pretext step, respectively. Therefore, one cannot reach the conclusion that Theorem 4 dominate Theorem 2. However, since we usually use neural networks to train processor f, which usually suffers from a large model capacity, Theorem 4 is better than Theorem 2 when the model capacity is large.

5 Experiments

In this section, we conduct experiments on synthetic data and real-world data, mainly to show that providing few downstream samples indeed hurt the model performance. Due to space limitations, we defer the experiment details to the supplementary materials.



(a) MSE under different n_0 (b) MSE under different λ (c) MSE under different d_f

Figure 1: **Algorithm performance under different hyperparameters on synthetic data.** (a) Our algorithm learns better with sufficient downstream samples (large n_0). MSE decreases to zero as downstream samples increase. (b) A large penalty forces the coarse representation to abandon the $\boldsymbol{y}$ information, leading to a large MSE. (c) When d_f is small, the model underfits; when d_f is large, the model suffers from a limited number of downstream samples.

5.1 Synthetic Data

We validate the statements we proved before in this section, showing that the processor-based learning fails when (a) extra downstream samples are insufficient or (b) the penalty parameter is too large.

Setup. We consider a linear regime, where $\boldsymbol{x}$ is generated from Gaussian distribution, $\boldsymbol{z} = \boldsymbol{x}[0 : d_z - 1] + \epsilon_{\boldsymbol{z}}$ and $\boldsymbol{y} = \boldsymbol{x}[0 : d_y - 1] + \epsilon_{\boldsymbol{y}}$, where $\epsilon_{\boldsymbol{z}}, \epsilon_{\boldsymbol{y}}$ are generated from Gaussian distributions. We run each experiment 5 times repeatedly and calculate its mean and standard deviation. We plot their 95% confidence bands in all the figures (light blue and light yellow). We left all the other details to the supplementary materials.

Algorithm and Metrics. We run our newly proposed algorithm (labeled as Ours), which first trains the processor and then does pretext and downstream tasks. We denote d_f as the coarse representation dimension and λ as the penalty coefficient in the loss (See Equation 1). We also run the standard self-supervised learning (labeled as SSL). In terms of metrics, we calculate the MSE on test downstream samples as the model performance. Intuitively, MSE is small when the features are learned well.

Analysis. We plot the results in Figure 1 and defer the specific statistics in the supplementary materials due to space limitations. Firstly, we plot the MSE as the number of extra samples n_0 varies from 30 to 140 in Figure 1(a). With large n_2, the downstream tasks benefit from small MSE, showing that the algorithm indeed finds the proper coarse representations (demonstrated in Theorem 1). Secondly, we test the case λ varies from 0.1 to 1.5, as plotted in Figure 1(b). When λ is large, the algorithm suffers from unsatisfying performances as demonstrated in Theorem 1. Thirdly, we test the case d_f varies from 1 to 12, and plot them in Figure 1(c).

5.2 Real-World Data (CIFAR-10)

To better show that the newly proposed processor-based method indeed fails in real tasks, we conduct experiments on CIFAR-10 [13] and choose rotation prediction [9] as the baseline SSL framework. The training set is split into 5k (labeled, extra data for processor training), 30k (unlabeled), and 15k (labeled) subsets without overlap to mimic the SSL training and a downstream classification task.

For the baseline, following [9], we train a plain rotation prediction task using 30k unlabeled training samples on a four-block NIN model [16]. Then we treat the 15k labeled training samples and 10k test samples as the training and test sets of the downstream classification task. A linear classifier is learned to conduct the task. For the processor-based learning, the only difference is that we now specify f in Eq. (1) as the first two blocks in a NIN model. When optimizing f, the 5k labeled and 30k unlabeled training samples are used to minimize Loss in Eq. (1). After obtaining f, we fix the weights of the first two blocks in the NIN model and go through the SSL training pipeline again, same as above.

n_0	1k	2k	3k	4k	5k	SSL
Full	9.95 (0.05)	9.97 (0.07)	9.99 (0.08)	10.01 (0.06)	9.96 (0.07)	79.31 (0.09)
Half	10.00 (0.07)	10.01 (0.09)	9.99 (0.08)	9.99 (0.08)	10.01 (0.09)	75.25 (0.04)

Table 1: **Failure on CIFAR-10.** Standard self-supervised learning can reach more than 75% top-1 accuracy on CIFAR-10, while the processor-based methods can only reach about 10% accuracy (nearly random guessing). Therefore, we conclude that training with insufficient downstream samples indeed hurt the feature extraction process, leading to unsatisfying model performances. Note that the statistics within the bracket stand for the 95% confidence band.

We test different model structures (labeled as *half* for a small model and *full* for a large model) and different number of extra data n_0 (from 1k to 5k) in Table 1. Table 1 illustrates that providing extra but insufficient data in advance hurts the performance (just like random guessing in practice).

6 Related Works

Self-Supervised Methods in Practice. There are three common approaches for Self-Supervised Learning (SSL): generative model based, contrastive learning based, and pretext based. Generative model based SSL [5–7] learns a bijective mapping between input and representation. Contrastive learning based SSL learns representations by maximizing the mutual information between the global and local features [1, 12, 19] or between the features of positive samples [3, 11, 24]. Pretext based SSL learns representations via handcrafted pretext tasks [8, 18, 20, 28]. These three approaches are quite different in technique, and we focus on studying the pretext based SSL in this paper.

Theory for Self-Supervised Learning. Although there are a number of great empirical works for SSL, the theoretical study of SSL is still at an early stage. The most related work to ours is given by Lee et al. [15]. They are the first to give a formulation of pretext-based SSL, and show that it can reduce the sample complexity of downstream tasks compared with supervised learning. They also point out that when the CI condition holds, the downstream sample complexity achieves the optimal, and it gets worse when the CI condition does not hold. In this paper, we further study the situation that the CI condition does not hold, and explore the idea of applying a learnable function to the input to make the CI condition hold. Other works [23, 25, 26] study the generalization error of contrastive learning based SSL, whose setting is different from our paper. Recently, Bansal et al. [2] analyze the generalization gap for most SSL methods. However, the rationality gap, which is a part of the generalization gap, cannot be theoretically bounded.

7 Conclusion and Future Work

In this work, we explore the idea of applying a learnable function f to the input to make conditional independence $f(\boldsymbol{x}) \perp \boldsymbol{z} \,|\, \boldsymbol{y}$ hold. We rigorously formulate the criteria that function f needs to satisfy. We design an ingenious loss function and prove that a function f minimizing the proposed loss satisfies the above criteria. We then theoretically study the number of extra downstream data required, by giving both model-free and model-dependent lower bounds of downstream sample size for learning the function f. We show that taking limited downstream data will hurt the performance of self-supervised learning. Numerical experiments on synthetic and real-world datasets both validate our theoretical results. In summary, our work partially provide evidence to validate the standard self-supervised learning in a sense by proving that involving additional but insufficient data will hurt downstream task performance.

The experiments have shown that under some proper hyper-parameters, taking downstream data to the representation learning phase can boost the pretext-based self-supervised learning. Thus, one possible future work is to analyze the upper bound of downstream sample size for learning the function f. Another possible direction could be exploring the idea of this paper for other self-supervised learning approaches, i.e., contrastive learning based approach.

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Supplementary Materials

We give all the proofs of lemmas and theorems here organized by theorems, i.e., Section A for Theorem 1 and related lemmas, Section B for Theorem 2, Section C for Theorem 3, and Section D for Theorem 4.

A Proof of Theorem 1

Theorem 1 (Rationality of Loss)**.** *Assume that there exists a ground truth $f^* \in \mathcal{F}$ satisfying both Criterion (C1) and (C2), where $\mathcal{F}$ is the candidate function class. Assume that for each $f \in \mathcal{F}$, the matrix $\mathbb{E}[f(\boldsymbol{x})f^\top(\boldsymbol{x})]$ is nonsingular. Let $\mathcal{A}_\mathcal{P}$ be the set of processors that minimize the population loss under data distribution $\mathcal{P}$:*

$$\mathcal{A}_\mathcal{P} = \left\{ f : f \in \arg\min_f \mathcal{L}(f; \mathcal{P}) \right\}.$$

Let $\mathcal{B}_\mathcal{P}$ be the set of processor that satisfy the criteria, namely:

$$\mathcal{B}_\mathcal{P} = \{ f : f \text{ satisfies Criterion (C1) and Criterion (C2)} \}.$$

Then by choosing a proper parameter λ, there exist a number of population distributions $\{\mathcal{P}\}$'s such that every function in $\mathcal{A}_\mathcal{P}$ satisfies Criterion (C1) and Criterion (C2), namely, the following defined set $\mathbb{S}$ is not empty:

$$\mathbb{S} \triangleq \{ \mathcal{P} : \mathcal{A}_\mathcal{P} \subset \mathcal{B}_\mathcal{P} \} \neq \phi.$$

To prove Theorem 1, we need to construct a data distribution $\mathcal{P}_0$ such that $\mathcal{P}_0 \in \mathbb{S}$. We construct such distribution as follows: Let $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$ be the input variables, pretext label, and downstream label, respectively. Firstly, let $\mathbb{E}[\boldsymbol{x}] = \mathbb{E}[\boldsymbol{y}] = \mathbb{E}[\boldsymbol{z}] = 0$ and $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] = I$. Given a marginal distribution of $(\boldsymbol{x}, \boldsymbol{y})$, construct a linear relationship $\boldsymbol{y} = B\boldsymbol{z}$ where the singular values of matrix B be $\sigma_1 = \cdots = \sigma_{d_y} = \sigma$. And choose the penalty $\lambda < \sigma^2$.

We first take a closer look at the loss function $\mathcal{L}(f) = \mathcal{L}_2 + \lambda\mathcal{L}_1$. We can rewrite it as $\mathcal{L}(f) = (1 - \frac{\lambda}{\sigma^2})\mathcal{L}_2 + \lambda(\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2)$. The first term is $\mathcal{L}_2$ multiplied by a coefficient, which still captures the information of $\boldsymbol{y}$ as $\mathcal{L}_2$ does. The second term intuitively captures the redundant information of $\boldsymbol{z}$.

We first provide Lemma A.1 and Lemma A.2 used during the proof of Theorem 1. Lemma A.1 shows that $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ is minimized if and only if all the redundant information of $\boldsymbol{z}$ is eliminated.

Lemma A.1. *Under the assumptions of Theorem 1 and data distribution $\mathcal{P}_0$ with penalty $\lambda < \sigma^2$, then $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ is minimized if and only if the conditional independence criterion (C1) holds, i.e., $\mathrm{Cov}[f(\boldsymbol{x}), \boldsymbol{z} \mid \boldsymbol{y}] = 0$.*

The advantage of decomposing $\mathcal{L}(f)$ into $(1 - \frac{\lambda}{\sigma^2})\mathcal{L}_2$ and $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ is that these two terms can be optimized individually. In other words, when $\mathcal{L}(f)$ is minimized, the above two terms are also minimized. We state it formally as the following Lemma A.2.

Lemma A.2. *Under the assumptions of Theorem 1 and data distribution $\mathcal{P}_0$ with penalty $\lambda < \sigma^2$, if $\mathcal{L}(f)$ is minimized, then $(1 - \frac{\lambda}{\sigma^2})\mathcal{L}_2$ and $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ are both minimized.*

According to Lemma A.2, for each function f that minimizes the loss, it also minimizes $(1 - \frac{\lambda}{\sigma^2})\mathcal{L}_2$ and $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ at the same time. For the first term, since $1 - \frac{\lambda}{\sigma^2}$ is a positive coefficient, $\mathcal{L}_2$ is minimized. Therefore, Criterion (C2) holds. Since the second term is minimized, Criterion (C1) holds by Lemma A.1. Thus, such f satisfies both Criterion (C1) and (C2).

We next prove Lemma A.1 and Lemma A.2 in Section A.1 and Section A.2, respectively,

A.1 Proof of Lemma A.1

Lemma A.1. *Under the assumptions of Theorem 1 and data distribution $\mathcal{P}_0$ with penalty $\lambda < \sigma^2$, then $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ is minimized if and only if the conditional independence criterion (C1) holds, i.e., $\mathrm{Cov}[f(\boldsymbol{x}), \boldsymbol{z} \mid \boldsymbol{y}] = 0$.*

Proof. Criterion (C1) is equivalent to

$$\Sigma_{f(\boldsymbol{x}),\boldsymbol{z}|\boldsymbol{y}} = \Sigma_{\boldsymbol{z}f(\boldsymbol{x})} - \Sigma_{\boldsymbol{y}f(\boldsymbol{x})}\Sigma_{\boldsymbol{y}\boldsymbol{y}}^{-1}\Sigma_{\boldsymbol{y}\boldsymbol{z}} = 0.$$

Notice that $\mathbb{E}[\boldsymbol{y}] = \mathbb{E}[\boldsymbol{z}] = \mathbb{E}[f(\boldsymbol{x})] = 0$, thus the above equation can be rewritten as

$$\mathbb{E}[f(\boldsymbol{x})\,\boldsymbol{z}^\top] = \mathbb{E}[f(\boldsymbol{x})\,\boldsymbol{y}^\top]\left(\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top]\right)^{-1}\mathbb{E}[\boldsymbol{y}\boldsymbol{z}^\top]. \tag{A.1}$$

On the other hand, we express the term $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ as

$$\begin{aligned}
&\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2 \\
&= \frac{1}{\sigma^2}\,\mathbb{E}\left\|\boldsymbol{y} - W^*_{\boldsymbol{y},f(\boldsymbol{x})}f(\boldsymbol{x})\right\|^2 - \mathbb{E}\left\|\boldsymbol{z} - W^*_{\boldsymbol{z},f(\boldsymbol{x})}f(\boldsymbol{x})\right\|^2 \\
&= \operatorname{tr}[\frac{1}{\sigma^2}\,\mathbb{E}\left[\left(\boldsymbol{y} - W^*_{\boldsymbol{y},f(\boldsymbol{x})}f(\boldsymbol{x})\right)\left(\boldsymbol{y} - W^*_{\boldsymbol{y},f(\boldsymbol{x})}f(\boldsymbol{x})\right)^\top\right] \\
&\quad - \mathbb{E}\left[\left(\boldsymbol{z} - W^*_{\boldsymbol{z},f(\boldsymbol{x})}f(\boldsymbol{x})\right)\left(\boldsymbol{z} - W^*_{\boldsymbol{z},f(\boldsymbol{x})}f(\boldsymbol{x})\right)^\top\right]] \\
&= \operatorname{tr}[\frac{1}{\sigma^2}\left(\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top] - \mathbb{E}\left[\boldsymbol{y}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{y}^\top\right]\right) \\
&\quad - \left(\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] - \mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]\right)] \\
&= \operatorname{tr}[\mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right] \\
&\quad - \frac{1}{\sigma^2}\,\mathbb{E}\left[\boldsymbol{y}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{y}^\top\right]] + \left(\frac{1}{\sigma^2}\,\mathbb{E}[\boldsymbol{y}^\top\boldsymbol{y}] - \mathbb{E}[\boldsymbol{z}^\top\boldsymbol{z}]\right) \\
&= \operatorname{tr}\left[\left(I - \frac{1}{\sigma^2}B^\top B\right)\mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]\right] + \left(\frac{1}{\sigma^2}\,\mathbb{E}[\boldsymbol{y}^\top\boldsymbol{y}] - \mathbb{E}[\boldsymbol{z}^\top\boldsymbol{z}]\right),
\end{aligned}$$

where the third equation holds because $W^*_{\boldsymbol{y},f(\boldsymbol{x})} = \mathbb{E}[\boldsymbol{y}f^\top(\boldsymbol{x})]\left(\mathbb{E}[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})]\right)^{-1}$ and the last equation follows the assumption that $\boldsymbol{y} = B\boldsymbol{z}$. Notice that the second term $\frac{1}{\sigma^2}\mathbb{E}\left[\boldsymbol{y}^\top\boldsymbol{y}\right] - \mathbb{E}\left[\boldsymbol{z}^\top\boldsymbol{z}\right]$ is unrelated to f. Therefore, minimizing $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ is equivalent to minimizing

$$T := \operatorname{tr}\left[\left(I - \frac{1}{\sigma^2}B^\top B\right)\mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}\left[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]\right]. \tag{A.2}$$

(a) We first prove the **sufficient condition**, namely, the conditional independence criterion (C1) leads to minimizing $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$.

Plugging Equation (A.1) to Equation (A.2), we have

$$\begin{aligned}
T &= \operatorname{tr}\left[\left(I - \frac{1}{\sigma^2}B^\top B\right)\mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}\left[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]\right] \\
&= \operatorname{tr}\left[\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]\left(I - \frac{1}{\sigma^2}B^\top B\right)\mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right]\left(\mathbb{E}\left[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})\right]\right)^{-1}\right] \\
&= \operatorname{tr}\left[\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{y}^\top\right]\left(\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top]\right)^{-1}\mathbb{E}[\boldsymbol{y}\boldsymbol{z}^\top]\left(I - \frac{1}{\sigma^2}B^\top B\right)\left(\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{y}^\top\right]\left(\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top]\right)^{-1}\mathbb{E}[\boldsymbol{y}\boldsymbol{z}^\top]\right)^\top\left(\mathbb{E}\left[f(\boldsymbol{x})\,f^\top(\boldsymbol{x})\right]\right)^{-1}\right].
\end{aligned}$$

By assumption that $\boldsymbol{y} = B\boldsymbol{z}$ and $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] = I$, we have

$$\begin{aligned}
&\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{y}^\top\right]\left(\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top]\right)^{-1}\mathbb{E}[\boldsymbol{y}\boldsymbol{z}^\top]\left(I - \frac{1}{\sigma^2}B^\top B\right)\left(\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{y}^\top\right]\left(\mathbb{E}[\boldsymbol{y}\boldsymbol{y}^\top]\right)^{-1}\mathbb{E}[\boldsymbol{y}\boldsymbol{z}^\top]\right)^\top \\
&= \mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]B^\top\left(\mathbb{E}[B\boldsymbol{z}\boldsymbol{z}^\top B^\top]\right)^{-1}B\,\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top]\left(I - \frac{1}{\sigma^2}B^\top B\right)\left(\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]B^\top\left(\mathbb{E}[B\boldsymbol{z}\boldsymbol{z}^\top B^\top]\right)^{-1}B\,\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top]\right)^\top \\
&= \mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]B^\top[BB^\top]^{-1}B\left(I - \frac{1}{\sigma^2}B^\top B\right)\left(\mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]B^\top[BB^\top]^{-1}B\right)^\top \\
&= \mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]B^\top[BB^\top]^{-1}\left[BB^\top - \frac{1}{\sigma^2}BB^\top BB^\top\right][BB^\top]^{-1}B\,\mathbb{E}\left[\boldsymbol{z}f(\boldsymbol{x})^\top\right] \\
&= \mathbb{E}\left[f(\boldsymbol{x})\,\boldsymbol{z}^\top\right]B^\top[BB^\top]^{-1}\cdot 0_{d_y\times d_y}\cdot[BB^\top]^{-1}B\,\mathbb{E}\left[\boldsymbol{z}f(\boldsymbol{x})^\top\right] \\
&= 0.
\end{aligned}$$

Furthermore, notice that the eigenvalues of $I - \frac{1}{\sigma^2} B^\top B$ is no less than zero by the definition of σ. And notice that the eigenvalues of $\mathbb{E}\left[\boldsymbol{z} f^\top(\boldsymbol{x})\right]\left(\mathbb{E}\left[f(\boldsymbol{x}) f^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right]$ is also no less than zero. Therefore, T reaches minimum when it reaches zero. To conclude, the sufficient condition holds.

(b) We next prove the **necessary condition**, namely, minimizing $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ leads to the conditional independence.

Under the assumption that there exists a ground truth f^* satisfying Criterion (C1) and (C2), we see from the sufficient condition that when T reaches the minimum, T must be equal to zero:

$$T = \operatorname{tr}\left[\left(I - \frac{1}{\sigma^2} B^\top B\right)\mathbb{E}\left[\boldsymbol{z} f^\top(\boldsymbol{x})\right]\left(\mathbb{E}\left[f(\boldsymbol{x}) f^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right]\right] = 0.$$

Since the matrix is semi-definite, we can omit the trace term and rewrite it as:

$$\mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right]\left(I - \frac{1}{\sigma^2} B^\top B\right)\mathbb{E}\left[\boldsymbol{z} f^\top(\boldsymbol{x})\right]\left(\mathbb{E}\left[f(\boldsymbol{x}) f^\top(\boldsymbol{x})\right]\right)^{-1} = 0.$$

Besides, since $\mathbb{E}\left[f(\boldsymbol{x}) f^\top(\boldsymbol{x})\right]$ is non-singular, we have

$$\mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right]\left(I - \frac{1}{\sigma^2} B^\top B\right)\mathbb{E}\left[\boldsymbol{z} f^\top(\boldsymbol{x})\right] = 0.$$

On the other hand, we represent the covariance as the follows based on the assumption $\mathbb{E}\left[\boldsymbol{z}\boldsymbol{z}^\top\right] = I$,

$$\Sigma_{f(\boldsymbol{x}),\boldsymbol{z}|\boldsymbol{y}} = \mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right] - \mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{y}^\top\right]\left(\mathbb{E}\left[\boldsymbol{y}\boldsymbol{y}^\top\right]\right)^{-1}\mathbb{E}\left[\boldsymbol{y}\boldsymbol{z}^\top\right] = \mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right]\left[I - \frac{1}{\sigma^2} B^\top B\right].$$

Finally, the covariance meets

$$\Sigma_{f(\boldsymbol{x}),\boldsymbol{z}|\boldsymbol{y}}\Sigma^\top_{f(\boldsymbol{x}),\boldsymbol{z}|\boldsymbol{y}} = \mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right]\left[I - \frac{1}{\sigma^2} B^\top B\right]\mathbb{E}\left[\boldsymbol{z} f(\boldsymbol{x})^\top\right] = 0.$$

This leads to the conclusion that $\Sigma_{f(\boldsymbol{x}),\boldsymbol{z}|\boldsymbol{y}} = 0$.

Combining (a) and (b), we finish the proof. □

A.2 Proof of Lemma A.2

Lemma A.2. *Under the assumptions of Theorem 1 and data distribution $\mathcal{P}_0$ with penalty $\lambda < \sigma^2$, if $\mathcal{L}(f)$ is minimized, then $(1 - \frac{\lambda}{\sigma^2})\mathcal{L}_2$ and $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ are both minimized.*

Proof. Notice that there exists a ground truth f^* that satisfies both Criterion (C1) and Criterion (C2). By definition of Criterion (C2), the ground truth f^* should minimize $\mathcal{L}_2$. On the other hand, we have proved in Section A.1 that satisfying Criterion (C2) leads to minimizing $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$.

Notice that

$$\mathcal{L}(f) = \lambda\mathcal{L}_1 + \mathcal{L}_2 = \left(1 - \frac{\lambda}{\sigma^2}\right)\mathcal{L}_2 + \lambda\left(\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2\right),$$

with $\lambda > 0$. Therefore, the ground truth f^* minimizes $\left(1 - \frac{\lambda}{\sigma^2}\right)\mathcal{L}_2$ and $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$ at the same time. Thus, any function f minimizing the loss must minimize both $\left(1 - \frac{\lambda}{\sigma^2}\right)\mathcal{L}_2$ and $\mathcal{L}_1 + \frac{1}{\sigma^2}\mathcal{L}_2$, or it would be larger than $\mathcal{L}(f^*)$. □

B Proof of Theorem 2

Theorem 2 (Model-Free Lower Bound)**.** *Assume that there exists a ground truth $f^* \in \mathcal{F}$ satisfying both Criterion (C1) and (C2), where $\mathcal{F}$ is the candidate function class. And assume that there exist function $f_1 \in \mathcal{F}$ and $f_2 \in \mathcal{F}$ such that covariance $\operatorname{Cov}[f_1(\boldsymbol{x}), \boldsymbol{y}] \neq 0$ and $\operatorname{Cov}[f_2(\boldsymbol{x}), \boldsymbol{z}] = 0$.*

Assume that for any $f_2 \in \mathcal{F}$, the matrix $\mathbb{E}[f(\boldsymbol{x})f^\top(\boldsymbol{x})]$ is nonsingular. Let $\mathcal{A}'_{\mathcal{P}}$ be the set of processor that minimizes the training loss with $n_0 = o(d_f)$ downstream samples and infinite pretext samples, where d_f is the dimension of $f(\boldsymbol{x})$,

$$\mathcal{A}'_{\mathcal{P}} = \left\{ f : f \in \arg\min_f \mathcal{L}_{\infty,n_0}(f;\mathcal{P}) \right\}.$$

And let $\mathcal{B}_{\mathcal{P}}, \mathbb{S}$ be defined as in Theorem 1. Then there exists *a distribution $\mathcal{P}^0 \in \mathbb{S}$ (which means $\mathcal{A}_{\mathcal{P}^0} \subset \mathcal{B}_{\mathcal{P}^0}$), such that all the function that minimizes the loss $\mathcal{L}_{\infty,n_0}$ cannot meet Criterion* (C1) *and* (C2) *simultaneously, namely:*

$$\mathcal{A}'_{\mathcal{P}^0} \cap \mathcal{B}_{\mathcal{P}^0} = \phi.$$

Proof. We consider the distribution $\mathcal{P}_0$ as in Section A, that is: Firstly, let $\mathbb{E}[\boldsymbol{x}] = \mathbb{E}[\boldsymbol{y}] = \mathbb{E}[\boldsymbol{z}] = 0$ and $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] = I$. Given a marginal distribution of $(\boldsymbol{x}, \boldsymbol{y})$, construct a linear relationship $\boldsymbol{y} = B\boldsymbol{z}$ where the singular values of matrix B be $\sigma_1 = \cdots = \sigma_{d_y} = \sigma$. And choose the penalty $\lambda < \sigma^2$.

Let us first consider the training process. We first claim that when $n_0 = o(d_f)$, for any f, the first term of $\mathcal{L}_{\infty,n_0}$ can be trained to zero, i.e.,

$$\frac{1}{n_0} \left\| Y^{(1)}_{down} - \widetilde{W}_2 f\left(X^{(1)}_{down}\right) \right\|^2 = 0,$$

by taking

$$\widetilde{W}_2 = Y^{(1)}_{down} f^\top\left(X^{(1)}_{down}\right) \left[f\left(X^{(1)}_{down}\right) f^\top\left(X^{(1)}_{down}\right) \right]^{-1},$$

where we abuse $f\left(X^{(1)}_{down}\right) \in \mathbb{R}^{d_f \times n_0}$.

Therefore, to minimize the loss $\mathcal{L}_{\infty,n'_2}$ only needs to minimize the second term of $\mathcal{L}_{\infty,n_0}$:

$$\hat{f} \in \arg\min_f \left[-\min_W \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} \|\boldsymbol{z} - Wf(\boldsymbol{x})\|^2 \right].$$

Note that for any f,

$$\min_W \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} \|\boldsymbol{z} - Wf(\boldsymbol{x})\|^2 = \mathbb{E}\|\boldsymbol{z}\|^2 - \mathrm{tr}\left[\mathbb{E}\left[\boldsymbol{z}f^\top(\boldsymbol{x})\right] \left(\mathbb{E}\left[f(\boldsymbol{x})f^\top(\boldsymbol{x})\right]\right)^{-1} \mathbb{E}\left[f(\boldsymbol{x})\boldsymbol{z}^\top\right] \right] \le \mathbb{E}\|\boldsymbol{z}\|^2.$$

When $f(\boldsymbol{x})$ is uncorrelated to $\boldsymbol{z}$ (i.e., $f = f_2$), the equality holds, i.e.,

$$\min_W \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} \|\boldsymbol{z} - Wf_2(\boldsymbol{x})\|^2 = \mathbb{E}\|\boldsymbol{z}\|^2.$$

Therefore, any function $\hat{f}$ that minimizes the loss satisfies

$$\min_W \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{z}} \left\|\boldsymbol{z} - W\hat{f}(\boldsymbol{x})\right\|^2 = \mathbb{E}\|\boldsymbol{z}\|^2.$$

We next prove that any $\hat{f}$ that minimizes training loss $\mathcal{L}_{\infty,n'_2}$ could not contain any information of $\boldsymbol{y}$. Under the condition that $\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]$ is invertible, we derive that

$$\begin{aligned}
\min_W \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\|\boldsymbol{y} - W\hat{f}(\boldsymbol{x})\right\|^2 &= \mathbb{E}\|\boldsymbol{y}\|^2 - \mathrm{tr}\left[\mathbb{E}\left[\boldsymbol{y}\hat{f}^\top(\boldsymbol{x})\right] \left(\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]\right)^{-1} \mathbb{E}\left[\hat{f}(\boldsymbol{x})\boldsymbol{y}^\top\right] \right] \\
&= \mathbb{E}\|\boldsymbol{y}\|^2 - \mathrm{tr}\left[B\,\mathbb{E}\left[\boldsymbol{z}\hat{f}^\top(\boldsymbol{x})\right] \left(\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]\right)^{-1} \mathbb{E}\left[\hat{f}(\boldsymbol{x})\boldsymbol{z}^\top\right] B^\top \right] \\
&= \mathbb{E}\|\boldsymbol{y}\|^2 - \mathrm{tr}\left[B^\top B\,\mathbb{E}\left[\boldsymbol{z}\hat{f}^\top(\boldsymbol{x})\right] \left(\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]\right)^{-1} \mathbb{E}\left[\hat{f}(\boldsymbol{x})\boldsymbol{z}^\top\right] \right] \\
&= \mathbb{E}\|\boldsymbol{y}\|^2 - \mathrm{tr}\left[B^\top B\right] \left(\mathbb{E}\|\boldsymbol{z}\|^2 - \min_W \mathbb{E}\left\|\boldsymbol{z} - W\hat{f}(\boldsymbol{x})\right\|^2 \right) \\
&= \mathbb{E}\|\boldsymbol{y}\|^2.
\end{aligned}$$

However, since there exists f_1 such that $\mathrm{Cov}(f_1(\boldsymbol{x}), \boldsymbol{y}) \neq 0$, leading to

$$\min_{W} \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\| \boldsymbol{y} - W\hat{f}\left(\boldsymbol{x}\right) \right\|^2 < \min_{W} \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\| \boldsymbol{y} - W f_1\left(\boldsymbol{x}\right) \right\|^2$$

Therefore, $\hat{f}$ violates Criterion (C2). The proof is done. □

C Proof of Theorem 3

Theorem 3 (Lower Bound for General Loss)**.** *Assume there exists a function f such that $f(\boldsymbol{x}) \perp \boldsymbol{z}$ holds, and assume $f(\boldsymbol{x}) \perp \boldsymbol{z}$ holds if and only if $\mathbb{E}\, g_1(\rho_1(\boldsymbol{z}, W^*_{\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x})))$ reaches the maximum. Assume that $g_2 {g_1}^{-1}$ is convex and non-decreasing, and assume there exists a linear transformation $V \in \mathbb{R}^{d_z \times d_z}$ such that*

$$\mathbb{E}[\rho_1(\bar{\boldsymbol{y}}, V\boldsymbol{z})] < \frac{1}{M} g_2 {g_1}^{-1} \max_f \mathbb{E}\left[g_1 \rho_1\left(W^*_{V\boldsymbol{z},f(\boldsymbol{x})} f(\boldsymbol{x}), V\boldsymbol{z}\right)\right] - \frac{1}{M} \min_f \mathbb{E}\left[g_2 \rho_2\left(W^*_{\boldsymbol{y},f(\boldsymbol{x})} f(\boldsymbol{x}), \boldsymbol{y}\right)\right],$$

where $M \geq {g_2}'(x), \forall x \in [0, \sup_{\boldsymbol{x},\boldsymbol{y}} \rho_2(W^*_{\boldsymbol{y},\hat{f}(\boldsymbol{x})} \hat{f}(\boldsymbol{x}), \boldsymbol{y}) + \sup_{\boldsymbol{y},\boldsymbol{z}} \rho_1(\bar{\boldsymbol{y}}, V\boldsymbol{z})]$, and $\bar{\boldsymbol{y}} = (\boldsymbol{y}, \vec{0}_{d_z - d_y})$ is the augmented vector of $\boldsymbol{y}$. Let $\mathcal{A}^g_{\mathcal{P}}$ be the set of processor that minimize the general training loss with $n_0 = o(d_f)$ downstream samples and infinite pretext samples, where d_f is the dimension of $f(\boldsymbol{x})$,

$$\mathcal{A}^g_{\mathcal{P}} = \left\{ f : f \in \arg\min_f \overline{\mathcal{L}}_{\infty,n_0}(f; \mathcal{P}) \right\}.$$

Let $\mathcal{B}_{\mathcal{P}}$ be defined as in Theorem 1. Then there exists *a distribution $\mathcal{P}^0$ such that all the function that minimizes the loss $\overline{\mathcal{L}}_{\infty,n_0}(f; \mathcal{P})$ cannot meet Criterion (C1) and (C2) simultaneously, namely:*

$$\mathcal{A}^g_{\mathcal{P}^0} \cap \mathcal{B}_{\mathcal{P}^0} = \phi.$$

Proof. Denote the trained predictor as $\hat{f}$. Notice that with $n = o\left(d_f\right)$, $\boldsymbol{y}_i = \widehat{W}_2 f\left(\boldsymbol{x}_i\right)$ by setting

$$\widehat{W}_2 = \left(Y^{(1)}_{down}\right)^\top \left[f\left(X^{(1)}_{down}\right) f\left(X^{(1)}_{down}\right)^\top \right]^{-1} f\left(X^{(1)}_{down}\right),$$

where we abuse the notation $f\left(X\right) \in \mathbb{R}^{n \times d_f}$. This leads to

$$\min_W g_2 \rho_2\left(Y^{(1)}_{down}, W\hat{f}\left(X^{(1)}_{down}\right)\right) = 0,$$

on the training set.

Therefore, $\hat{f}$ is trained to minimize the loss

$$\mathcal{L}_1 = -\min_W \mathbb{E}\, g_1 \rho_1\left(\boldsymbol{z}, W f\left(\boldsymbol{x}\right)\right),$$

leading to the independence between $f(\boldsymbol{x})$ and $\boldsymbol{z}$.

By the definition of ρ_2 and ρ_1, we have for any $V > 0$,

$$\min_{W_1} \mathbb{E}\, g_2 \rho_2\left(\boldsymbol{y}, W_1 f\left(\boldsymbol{x}\right)\right) \triangleq \mathbb{E}\, g_2 \rho_2\left(\boldsymbol{y}, \widehat{W}_1 \hat{f}\left(\boldsymbol{x}\right)\right) = \mathbb{E}\, g_2 \rho_1\left(\bar{\boldsymbol{y}}, \widehat{\overline{W}}_1 \hat{f}\left(\boldsymbol{x}\right)\right), \tag{A.3}$$

where $\bar{\boldsymbol{y}}, \widehat{\overline{W}}_1$ are the augmented version filled with zero. By the upper bound of $g'\left(\cdot\right)$, we have

$$\begin{aligned}
& g_2 \rho_1\left(\bar{\boldsymbol{y}}, \widehat{\overline{W}}_1 \hat{f}\left(\boldsymbol{x}\right)\right) - g_2 \rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}\left(\boldsymbol{x}\right)\right) \\
=& {g_2}'\left(\xi\right) \left[\rho_1\left(\bar{\boldsymbol{y}}, \widehat{\overline{W}}_1 \hat{f}\left(\boldsymbol{x}\right)\right) - \rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}\left(\boldsymbol{x}\right)\right)\right] \\
\geq& {g_2}'\left(\xi\right) \left[-\rho_1\left(\bar{\boldsymbol{y}}, V\boldsymbol{z}\right)\right] \\
\geq& -M \rho_1\left(\bar{\boldsymbol{y}}, V\boldsymbol{z}\right).
\end{aligned} \tag{A.4}$$

where the first equation is due to mean value theorem of integrals with point ξ. The second equality is due to ${g_2}'(\xi) \geq 0$ and triangle inequality. And the third inequality is due to the condition ${g_2}'(\xi) \leq M$, and $\xi \leq \max\{\rho_1\left(\bar{\boldsymbol{y}}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right), \rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right)\} \leq Q_1 + Q_2$.

Besides, notice that

$$
\begin{aligned}
&\mathbb{E}\, g_2 \rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right) \\
=&\mathbb{E}\, g_2 {g_1}^{-1} g_1\left(\rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right)\right) \\
\overset{(i)}{\geq}& g_2 {g_1}^{-1}\, \mathbb{E}\, g_1\left(\rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right)\right) \\
\overset{(ii)}{\geq}& g_2 {g_1}^{-1} \min_W \mathbb{E}\, g_1\left(\rho_1\left(V\boldsymbol{z}, W \hat{f}(\boldsymbol{x})\right)\right) \\
=& g_2 {g_1}^{-1} \max_f \min_W \mathbb{E}\, g_1\left(\rho_1\left(V\boldsymbol{z}, W f(\boldsymbol{x})\right)\right),
\end{aligned}
\tag{A.5}
$$

where inequality (i) follows Jensen's inequality with condition $g_2 {g_1}^{-1}$ is convex, and inequality (ii) follows that $g_2 {g_1}^{-1}$ is increasing. The final equation is due to the independence between $f(\boldsymbol{x})$ and $\boldsymbol{z}$.

Combining Equation (A.3), Equation (A.4), and Equation (A.5), leads to

$$
\begin{aligned}
&\min_{W_1} \mathbb{E}\, g_2 \rho_2\left(\boldsymbol{y}, W_1 f(\boldsymbol{x})\right) \\
=&\mathbb{E}\, g_2 \rho_1\left(\bar{\boldsymbol{y}}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right) \\
\geq&\mathbb{E}\, g_2\left(\rho_1\left(V\boldsymbol{z}, \widehat{\overline{W}}_1 \hat{f}(\boldsymbol{x})\right)\right) - M\, \mathbb{E}\, \rho_1\left(\bar{\boldsymbol{y}}, V\boldsymbol{z}\right) \\
\geq& g_2 {g_1}^{-1} \max_f \min_W \mathbb{E}\, g_1\left(\rho_1\left(V\boldsymbol{z}, W \hat{f}(\boldsymbol{x})\right)\right) - M\, \mathbb{E}\, \rho_1\left(\bar{\boldsymbol{y}}, V\boldsymbol{z}\right) \\
>& \min_f \min_W \mathbb{E}\, g_2 \rho_2\left(\boldsymbol{y}, W_1 f(\boldsymbol{x})\right).
\end{aligned}
$$

This contradicts with Criterion (C2). The proof is done. □

D Proof of Theorem 4

Theorem 4 (Model-Dependent Lower Bound)**.** *Assume that the function class of the processor can be decomposed as $\mathcal{F} = \mathcal{F}_1 \times \mathcal{F}_2$, where there exists a function $f_0 \in \mathcal{F}$ such that $\mathrm{Cov}[f_0(\boldsymbol{x}), \boldsymbol{y}] \neq 0$ and a function $f_2 \in \mathcal{F}_2$ such that $f_2(\boldsymbol{x}) \perp \boldsymbol{z}$. And assume matrix $\mathbb{E}[f(\boldsymbol{x}) f^\top(\boldsymbol{x})]$ is nonsingular for all $f \in \mathcal{F}$. Let $\mathcal{A}'_{\mathcal{P}}$ be the set of processor that minimize the training loss with $n_0 = o(\mathcal{M}(\mathcal{F}_1, \mathcal{L}))$ downstream samples and infinite pretext samples, where d_f is the dimension of $f(\boldsymbol{x})$,*

$$
\mathcal{A}'_{\mathcal{P}} = \left\{ f : f \in \arg\min_f \mathcal{L}_{\infty, n_0}(f; \mathcal{P}) \right\}.
$$

And let $\mathcal{B}_{\mathcal{P}}, \mathbb{S}$ be defined as in Theorem 1. Then there exists *a distribution $\mathcal{P}^0 \in \mathbb{S}$ (which means $\mathcal{A}_{\mathcal{P}^0} \subset \mathcal{B}_{\mathcal{P}^0}$), such that all the function that minimizes the loss $\mathcal{L}_{\infty, n_0}$ cannot meet Criterion* (C1) *and* (C2) *simultaneously, namely:*

$$
\mathcal{A}'_{\mathcal{P}^0} \cap \mathcal{B}_{\mathcal{P}^0} = \phi.
$$

We consider the distribution $\mathcal{P}_0$ as in Section A, that is: Firstly, let $\mathbb{E}[\boldsymbol{x}] = \mathbb{E}[\boldsymbol{y}] = \mathbb{E}[\boldsymbol{z}] = 0$ and $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] = I$. Given a marginal distribution of $(\boldsymbol{x}, \boldsymbol{y})$, construct a linear relationship $\boldsymbol{y} = B\boldsymbol{z}$ where the singular values of matrix B be $\sigma_1 = \cdots = \sigma_{d_y} = \sigma$. And choose the penalty $\lambda < \sigma^2$.

Firstly, consider the training process. Under the assumptions on $\mathcal{F}_1 \times \mathcal{F}_2$, there exists a $f_q(f_2(\cdot))$ such that:

(a) on the one hand, $f_q(f_2(\cdot))$ minimize $\mathcal{L}_1$.

$$
f_q(f_2(\cdot)) \in \arg\min_f \mathcal{L}_1(f).
$$

Note that $f_q f_2(\boldsymbol{x}) \perp \boldsymbol{z}$ follows $f_2(\boldsymbol{x}) \perp \boldsymbol{z}$. As a result, due to the assumption $\mathbb{E}\,\boldsymbol{z} = 0$, we have

$$\arg\min_W \|\boldsymbol{z} - W f_q f_2(\boldsymbol{x})\| = 0,\ \min_W \|\boldsymbol{z} - W f_q f_2(\boldsymbol{x})\|^2 = \|\boldsymbol{z}\|^2.$$

And further notice that for any predictor f, we have

$$\min_W \|\boldsymbol{z} - W f(\boldsymbol{x})\|^2 \leq \|\boldsymbol{z} - 0 f(\boldsymbol{x})\|^2 = \|\boldsymbol{z}\|^2,$$

therefore, $f_q(f_2(\cdot))$ minimize $\mathcal{L}_1$.

(b) on the other hand, $f_q(f_2(\cdot))$ makes $\mathcal{L}_2$ equal to zero on the training set. Due to the definition of model capacity, when $n < o(\mathcal{M}(\mathcal{F}_1, \mathcal{L}))$, by fixing f_2, there always exists f_q such that it can fit any n samples:

$$\min_W \left\| Y_{down}^{(1)} - W f_q(f_2(\boldsymbol{x}_i)) \right\|^2 = 0.$$

As a result, $f_q(f_2(\cdot))$ can minimize the training loss since it minimize both $\mathcal{L}_1$ and $\mathcal{L}_2$. Therefore, any predictor $\hat{f}(\cdot)$ that minimize the training loss must minimize $\mathcal{L}_1$, leading to

$$\min_W \mathbb{E}\,\|\boldsymbol{z} - W \hat{f}(\boldsymbol{x})\|^2 = \mathbb{E}\,\|\boldsymbol{z}\|^2.$$

Next we consider the predictor $\hat{f}(\cdot)$.

$$\begin{aligned}
&\min_W \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\| \boldsymbol{y} - W \hat{f}(\boldsymbol{x}) \right\|^2 \\
=& \mathbb{E}\left(\|\boldsymbol{y}\|^2\right) - \operatorname{tr}\left[\mathbb{E}\left(\boldsymbol{y}\hat{f}^\top(\boldsymbol{x})\right)\left(\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left(\hat{f}(\boldsymbol{x})\boldsymbol{y}^\top\right)\right] \\
=& \mathbb{E}\left(\|\boldsymbol{y}\|^2\right) - \operatorname{tr}\left[B\,\mathbb{E}\left(\boldsymbol{z}\hat{f}^\top(\boldsymbol{x})\right)\left(\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left(\hat{f}(\boldsymbol{x})\boldsymbol{z}^\top\right)B^\top\right] \\
=& \mathbb{E}\left(\|\boldsymbol{y}\|^2\right) - \operatorname{tr}\left[B^\top B\,\mathbb{E}\left(\boldsymbol{z}\hat{f}^\top(\boldsymbol{x})\right)\left(\mathbb{E}\left[\hat{f}(\boldsymbol{x})\hat{f}^\top(\boldsymbol{x})\right]\right)^{-1}\mathbb{E}\left(\hat{f}(\boldsymbol{x})\boldsymbol{z}^\top\right)\right] \\
=& \mathbb{E}\left(\|\boldsymbol{y}\|^2\right) - \operatorname{tr}\left[B^\top B\left(\min_W \|\boldsymbol{z} - W\hat{f}(\boldsymbol{x})\|^2 - \|\boldsymbol{z}\|^2\right)\right] \\
=& \mathbb{E}\left(\|\boldsymbol{y}\|^2\right).
\end{aligned}$$

Notice that $\min_W \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}} \|\boldsymbol{y} - W f_1(\boldsymbol{x})\|^2 < \mathbb{E}\left(\|\boldsymbol{y}\|^2\right)$ since $\operatorname{Cov}[f_1(\boldsymbol{x}), \boldsymbol{y}] \neq 0$. Therefore,

$$\mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\| \boldsymbol{y} - W^*_{\boldsymbol{y},\hat{f}(\boldsymbol{x})} \hat{f}(\boldsymbol{x}) \right\|^2 > \min_f \mathop{\mathbb{E}}_{\boldsymbol{x},\boldsymbol{y}} \left\| \boldsymbol{y} - W^*_{\boldsymbol{y},f(\boldsymbol{x})} f(\boldsymbol{x}) \right\|^2$$

which contradicts to the Criterion (C2).

E Illustration of redundant information

In the main text, we state that the processor-based SSL aims at eliminating the redundant information. For better understanding, let us consider a simple case. Suppose $\boldsymbol{x}$ is a d_x-dimensional random vector. Let $\boldsymbol{z}$ and $\boldsymbol{y}$ consist of the first d_z and d_y elements of $\boldsymbol{x}$, respectively, i.e., $\boldsymbol{z} = \boldsymbol{x}[0:d_z-1]$ and $\boldsymbol{y} = \boldsymbol{x}[0:d_y-1]$. When we execute the standard self-supervised learning (with linear representation $\hat{\psi}$) using infinite pretext samples, the learned representation has rank d_z which consists of d_y-dimensional useful information and $(d_z - d_y)$-dimensional redundancy information that we cannot eliminate (See Figure 2). Therefore, we need $\tilde{\mathcal{O}}(d_z)$ samples to train the liner layer $\widehat{W}$ in the downstream tasks. In comparison, the processor approach requires a function f that eliminates the redundancy information at the beginning, i.e., $f(\boldsymbol{x}) = \boldsymbol{x}[0:d_y-1, d_z:d_x]$. Hopefully, $f(\boldsymbol{x})$ will consist of all the useful information and some information of $\boldsymbol{x}[d_z:d_x-1]$, without any information of $\boldsymbol{x}[d_y:d_z-1]$. And then we apply pretext task on $(f(\boldsymbol{x}), \boldsymbol{z})$, which further removes the information of $\boldsymbol{x}[d_z:d_x-1]$. Thus, the rank of representation $\widehat{\psi}(\boldsymbol{x})$ could be reduced to d_y. In this way, we will only need $\tilde{\mathcal{O}}(d_y)$ samples for the downstream task.

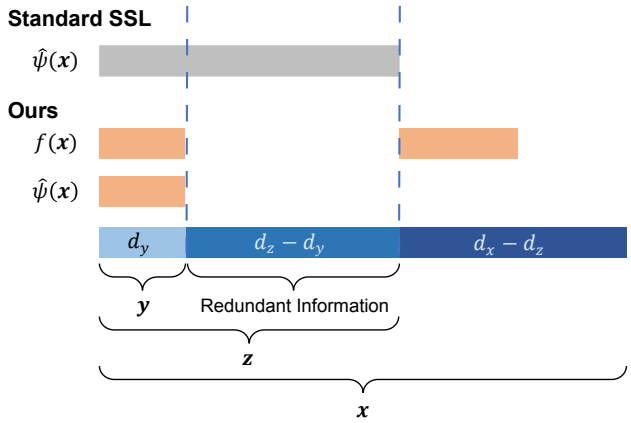

Figure 2: A simple case for understanding redundant information. Here $\boldsymbol{y}$ is the prefix of $\boldsymbol{z}$, and $\boldsymbol{z}$ is the prefix of $\boldsymbol{x}$. For the standard SSL, the learned representation $\widehat{\psi}(\boldsymbol{x})$ consists of not only the information of $\boldsymbol{y}$ but also the redundant information in $\boldsymbol{z}$. For the processor-based SSL, the learned coarse representation $f(\boldsymbol{x})$ contains the information of $\boldsymbol{y}$ and some information of $\boldsymbol{x}$, and does not include any redundant information in $\boldsymbol{z}$. Then the final learned representation $\widehat{\psi}(\boldsymbol{x})$ will only contain the information of $\boldsymbol{y}$, hopefully.

F Experimental details

F.1 Experimental details for synthetic data

Datasets and environment. The dataset is simulated as follows. Firstly, we randomly generate n_1 pretext samples $\{(x^i_{pre}, z^i_{pre})\}_{i\in[n_1]}$, where $x^i_{pre} \in \mathbb{R}^{d_x}$ is generated from Gaussian distribution $\mathcal{N}(0, I_{d_x})$, and $z^i_{pre} \in \mathbb{R}^{d_z}$ is a perturbation of the first d_z elements of x^i_{pre}, namely, $z^i_{pre} = x^i_{pre}[0 : d_z] + 0.01 \cdot \varepsilon^i_1$, where $\varepsilon^i_1 \sim \mathcal{N}(0, I_{d_z})$. Similarly, we generate n_2 downstream samples $\{(x^i_{down}, y^i_{down})\}_{i\in[n_2]}$, where $x^i_{down} \in \mathbb{R}^{d_x}$ is generated from Gaussian distribution $\mathcal{N}(0, I_{d_x})$ and $y^i_{down} \in \mathbb{R}^{d_y}$ is a perturbation of the first d_y elements of x^i_{down}, i.e., $y^i_{down} = x^i_{down}[0{:}d_y] + 0.01 \cdot \varepsilon^i_2$, where $\varepsilon^i_2 \sim \mathcal{N}(0, I_{d_y})$.

Similarly, we generate n_2 downstream samples $\{(x^i_{down}, y^i_{down})\}_{i\in[n_2]}$, where $x^i_{down} \in \mathbb{R}^{d_x}$ is generated from Gaussian distribution $\mathcal{N}(0, I_{d_x})$ and $y^i_{down} \in \mathbb{R}^{d_y}$ is a perturbation of the first d_y elements of x^i_{down}, i.e., $y^i_{down} = x^i_{down}[0{:}d_y] + 0.01 \cdot \varepsilon^i_2$, where $\varepsilon^i_2 \sim \mathcal{N}(0, I_{d_y})$.

Besides, we generate n_0 extra samples $\{(x^i_{extra}, y^i_{extra})\}_{i\in[n_0]}$, where $x^i_{extra} \in \mathbb{R}^{d_x}$ is generated from Gaussian distribution $\mathcal{N}(0, I_{d_x})$ and $y^i_{extra} \in \mathbb{R}^{d_y}$ is a perturbation of the first d_y elements of x^i_{extra}, i.e., $y^i_{extra} = x^i_{extra}[0{:}d_y] + 0.01 \cdot \varepsilon^i_3$, where $\varepsilon^i_3 \sim \mathcal{N}(0, I_{d_y})$. We generate n_t test samples under the same procedure.

In each run of our proposed algorithm, we first use n_0 extra samples and n_1 pretext samples to train the processor. Then we apply the processor as well as the n_1 pretext samples, n_2 downstream samples to finish the training. For a fair comparison, during the SSL procedure, we use n_1 pretext samples to train the feature and then use $n_2 + n_0$ downstream samples to learn the linear layer. We run each experiment 5 times repeatedly and calculate its mean and standard deviation. We plot their 95% confidence bands in all the figures.

The rationality of the loss $\mathcal{L}(f)$. Set $d_x = 100$, $d_z = 80$, $d_y = 5$, $d_f = 5$, $n_1 = 20000$ and $\lambda = 0.1$. We plot the MSE as n_2 varies from 30 to 140 in Figure 1(a). With large n_2, the downstream tasks benefit from small MSE, showing that the algorithm indeed finds the proper coarse representations. Theorem 1 demonstrates this phenomenon, which claims that with sufficient downstream task samples, the proposed algorithm finds the proper coarse representations.

We further remark that in the case $n_2 = 80$, the SSL suffers from a large MSE. It is related to the "double decent" phenomenon in the downstream tasks, and MSE reaches the maximum when $n_2 = d_z$.

We finally remark that when $n_2 \in [40, 80]$, the proposed algorithm outperforms SSL in terms of MSE.

Large coefficient λ harms the performance. Set $d_x = 100$, $d_z = 80$, $d_y = 5$, $d_f = 5$, $n_1 = 20000$ and $n_2 = 30$. We test the case λ varies from 0.1 to 1.5, as plotted in Figure 1(b). Figure 1(b) illustrates that when λ is large, the algorithm suffers from unsatisfying performances. Theorem 1 demonstrates the phenomenon. Intuitively, that is because $\boldsymbol{z}$ contains the information of $\boldsymbol{y}$. With a large coefficient λ, the trained classifier tends to eliminate the information of y, leading to a lousy model performance (MSE).

Large dimension d_f harms the performance. Set $d_x = 100$, $d_z = 80$, $d_y = 5$, $n_1 = 20000$, $n_2 = 30$, and $\lambda = 0.1$. We test the case d_f varies from 1 to 12, and plot them in Figure 1(c). Firstly, notice that MSE reaches the minimum when $d_f = d_y$, that is because we force the coarse representation to have no redundancy information by limiting its dimension. Secondly, notice that the figure shows a Basin-type phenomenon and reaches the minimum when $d_f = d_y$. On the one hand, when $d_f < d_y$, the coarse representation is forced to lose the information of y, leading to the underfitting phenomenon. On the other hand, when $d_f > d_y$, the performance of coarse representation is limited by the downstream sample numbers, as illustrated in Theorem 2 and Theorem 4.

F.2 Experimental details for real-world data

All experiments can be handled with an RTX 2080Ti graphic card.

Data split strategy. We first classify the samples according to their categories and sort them according to their index in the CIFAR-10 dataset. And then pick the correct number of samples in each class one by one without overlapping. For each subset, we assure that there is the same number of samples for each class.

Model training. For standard SSL training, we directly follow all the hyperparameters and training strategies from [9] except for the training set size.

For the processor-based learning, we also copy the hyperparameters from the standard setting, which is necessary. There are two additional SGD optimizers for training $W^*_{\boldsymbol{y},f(\boldsymbol{x})}$ and $W^*_{\boldsymbol{z},f(\boldsymbol{x})}$, which are two linear layers with the same input size and different output size in our implementation. We set the learning rate as 0.1 and train for 20 times before calculating the loss function used to update f. When optimizing Equation (1), λ is set to 0.1 to balance the two loss terms in Equation (1).

For the half NIN model, we half the channel number in each block. Consequently, the size of all linear layers should also be changed.